

Title Software Engineering –

Task 2. Dairy Product Supply System

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Submitted by Wycliffe Bahati

Matriculation Number: 92014201

Tutor: Prof. Dr. Holger Klus

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Table of Contents

Contents

Project Proposal..... 3

Requirements 3

 Functional requirements..... 3

 Non-Functional requirements 4

Use Case Diagram 5

Domain Model Diagram..... 6

Risks 6

Glossary 7

Project Proposal

A dairy cooperative would like to implement a new system that covers the supply chain of milk from milk collection from farmers to selling to the customers. The system will provide real-time data, enhance traceability, improve quality control, and facilitate efficient communication among all stakeholders.

Farmers will be able to input production data and view payment status for the milk supplied to the co-operative. The management should be able to view the inventory and orders for milk in real time. Distribution department should be able to track delivery status of the orders and route information. Retailers will place orders and track order status and manage sales. End customers should be able to order milk, track delivery and view milk quality status. Quality of the milk is recorded hourly and at the end of the day quality report is generated. The system should be able to handle sudden increase in amount of milk or customers without performance degradation. There should be restrict on accessing information and if possible. The system should have a user-friendly interface and should have high availability. An administrator will be responsible to add users and manage access role on the system

In this project, the requirements might evolve, the development process will be interactive involving the stakeholders. Due to this specification Agile software development methodologies will be used.

Requirements

Functional requirements

Farmers

- Input production data for milk
- View Payment status

The management

- View milk inventory
- View orders for milk real time

Distribution department

- Track delivery status
- Maintain routes

Quality Manager

- Measure and record milk quality information hourly

Admin

- Manage user roles, and access control.

Inventory manager

- Manage inventory levels

Non-Functional requirements

Scalability

- The system should handle increasing amounts of data and users without performance degradation.

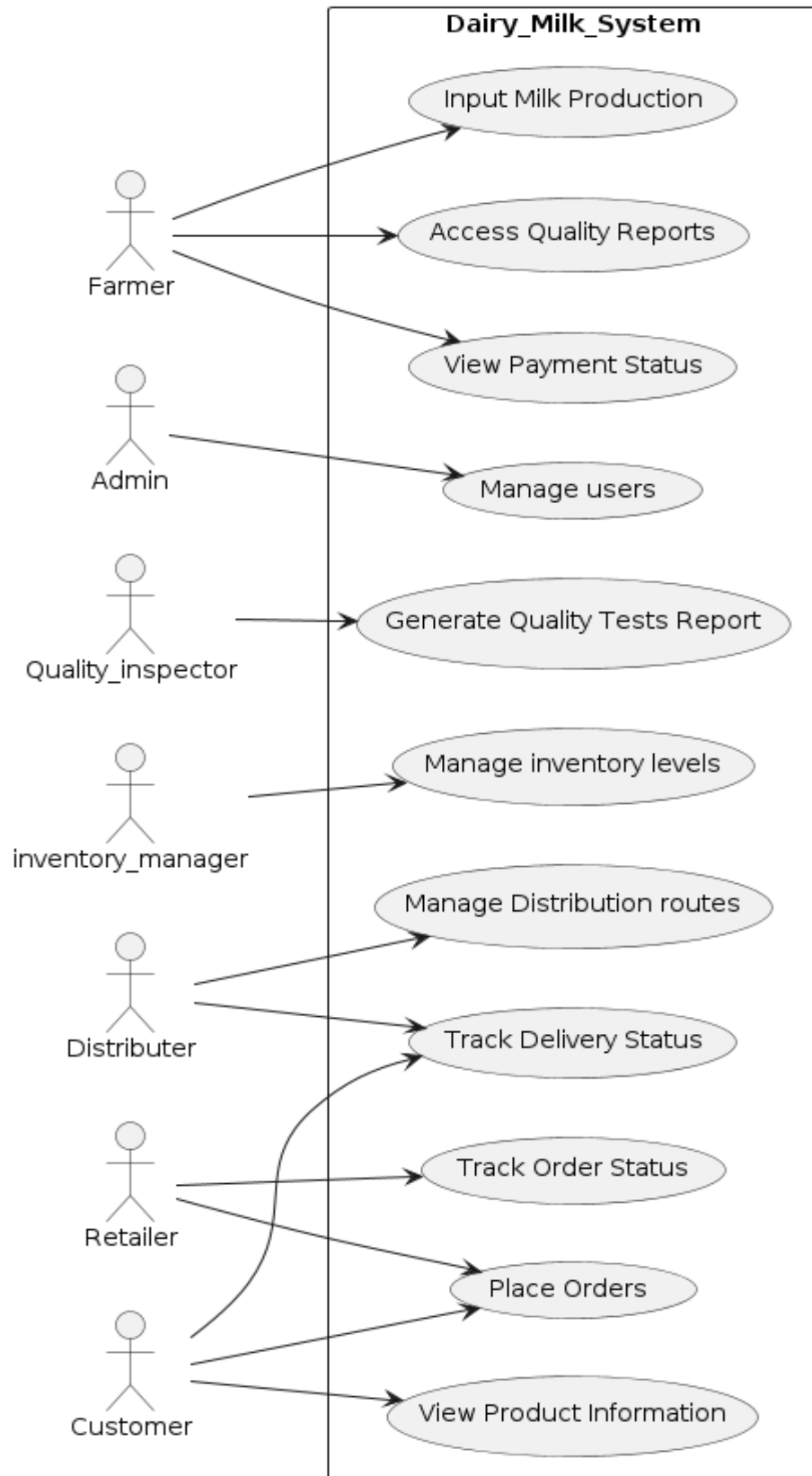
Security

- Role-based access control to ensure that users only access data relevant to their roles.
- Data encryption for sensitive information.

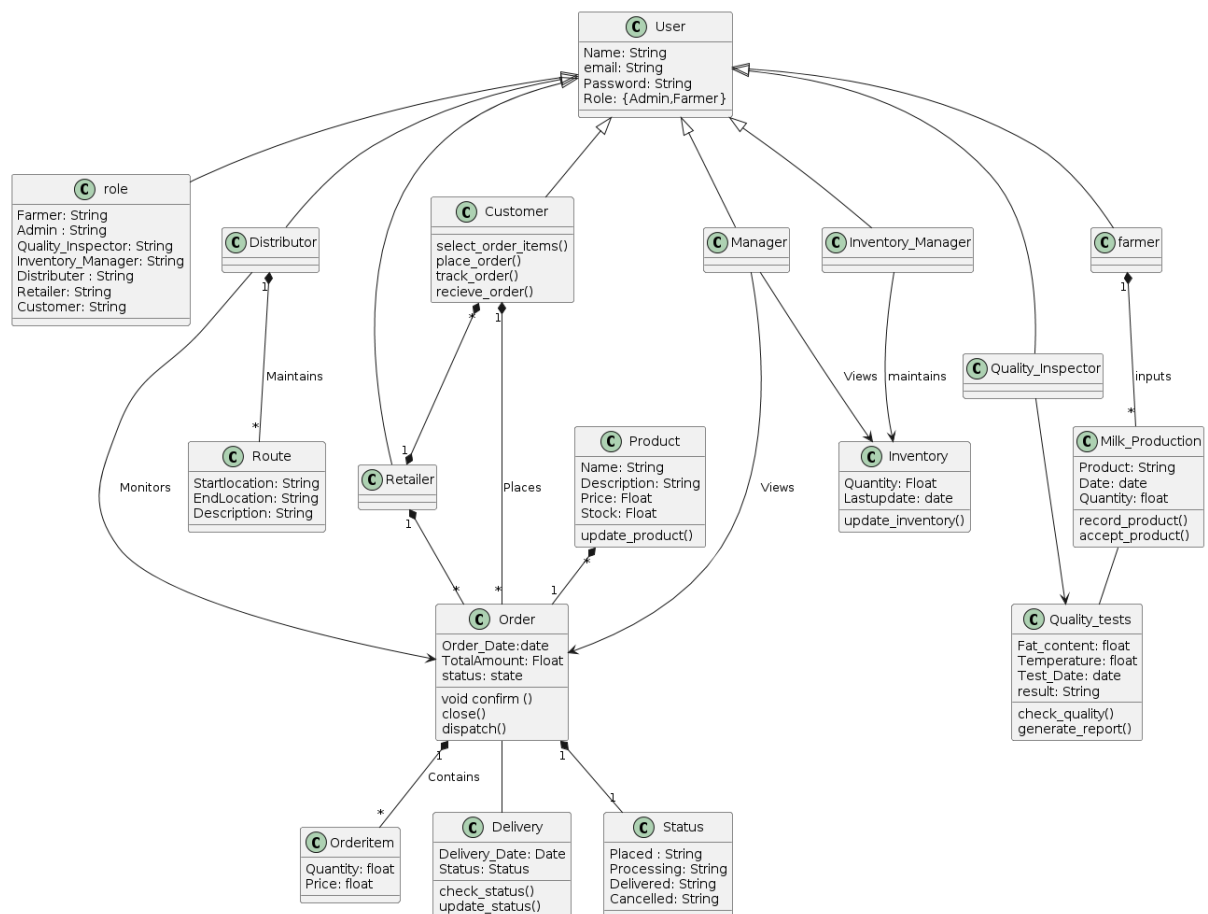
Usability

- User-friendly interfaces for all user roles

Use Case Diagram



Domain Model Diagram



Risks

Technical challenges

Team is using new technology that they are not familiar with

Mitigation

Rely on a lot of research and past projects to minimize the exposure to derail the project to completion

Time Management

Involvement in other daily activities might not allow enough time to complete the project on time.

Mitigation

Break the project into independent module and implement one module at a time to complete

Glossary

Farmer – Person responsible for milk production and supply to the Dairy shop. responsible for submitting the supplied products

Distributor - Person responsible for the supply of milk the retailers and maintain route information

Customer – End user who consumes the milk produced and supplied by the system.

Product - Different types of milk supplied by the farmer. Example goat milk or cow milk

Manager – User responsible for the overall health of the dairy. Oversea the supply and distribution of milk and payment to farmers

Inventory Manager – User responsible for keeping stock of how much milk is available for distribution

Quality inspector – User responsible for the quality of the milk supplied and distribution by the dairy.

Inventory – Stock available per product at a given date and time

Route – Contains destination that the dairy products is supplied. Contains both source and destination.

Retailer – Acts as a middleman between the Dairy shop and the Consumer. The Distributor supplied the milk products to retailers

Quality test – Parameters which the dairy products must meet to be accepted by the dairy shop.

Order – Contains 1 or more order items that are placed by the consumer or retailer

Order item – Represent the individual product that can be ordered by the consumer or retailer

Delivery – Contains status of an order placed by the consumer or retailer